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Unit 2 : Session 3 : SLO 1 | Advantages and Disadvantages of an Array

I. Advantages Fill in the Blanks

1. direct (random/constant-time) index-based $O(1)$ access.
2. contiguous elements sit in contiguous memory locations.
3. fixed (large) good for storing a fixed number of similar-typed items.
4. fixed size is fixed once declared.
5. repetitive (iterative) loops make bulk operations easy.
6. overhead (need) no need to declare many separate variables.
7. known (exact/sufficient) memory is efficient when the size matches the need.
8. constant ($O(1)$) index lookup time.

II. Disadvantages Fill in the Blanks

1. compile (declaration) size must be known at compile time (for static arrays).
2. fixed size cannot change at runtime.
3. shifting middle insertion requires shifting other elements.
4. shifting deletion also involves shifting elements.
5. too large (over-declared) over-declaring wastes memory.
6. same only one data type per array.
7. $O(n)$ (linear) worst-case search time in an unsorted array.